DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 1: Assignment

Financial Markets · Time Value of Money · Yield Curves

Instructions

- This assignment covers Lectures 1–3 and the Week 1 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script or notebook cell; submit your code alongside your written answers.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Financial Markets: Concepts (25 marks)

A1. **(4 marks)** A friend claims: "Government bonds are completely risk-free, so a smart investor should hold nothing else." Identify *two* distinct risks a government-bond-only investor still bears, and explain each in 2–3 sentences.

A2. **(6 marks)** Classify each instrument below as (i) a fixed/creditor claim, (ii) a residual/equity-like claim, or (iii) a derivative. Justify each classification in one sentence.

(a) A recurring bank deposit.
(b) A partnership stake in a two-person consulting firm.
(c) A rainfall-indexed crop insurance contract paying out if monsoon rainfall is below a threshold.
(d) A preference share paying a fixed dividend, but only after all bank loans are serviced.
(e) An interest rate swap exchanging a fixed rate for a floating (MIBOR-linked) rate.
(f) A zero-coupon bond purchased at a discount to face value.

A3. **(5 marks)** Explain, in your own words and *without* using any pricing formula, why a bond's market price must fall when prevailing market interest rates rise. Your answer should mention what happens to *newly issued* bonds and why that affects the attractiveness of an *existing* bond.

A4. **(5 marks)** A company needs to raise Rs. 200 crore for expansion. It is considering either issuing new bonds or issuing new equity. From the point of view of the company's *existing* shareholders, give one advantage and one disadvantage of each financing choice (four points total).

A5. **(5 marks)** A call option and a forward contract both give exposure to a rise in the underlying's price. Explain the key economic difference between them, referencing (i) whether a premium is paid upfront, and (ii) the shape of each contract's payoff diagram below vs. above the strike/delivery price.

Part B – Time Value of Money (35 marks)

B1. **(5 marks)** Rs. 2,50,000 is invested for 15 years. Compute the future value under: (a) simple interest at 9% p.a., (b) annual compound interest at 9% p.a., (c) continuous compounding at 9% p.a. Comment briefly on the pattern you observe.

B2. **(5 marks)** A bank advertises a personal loan at a nominal rate of 18% p.a., compounded monthly. Compute the effective annual rate (EAR). If a competing bank offers a flat 19.5% p.a. EAR with no compounding subtlety, which loan is cheaper for the borrower? Show your comparison.

B3. **(8 marks)** Derive, from first principles (i.e., by summing the geometric series – do not simply quote the final formula), the present value of an ordinary annuity paying A at the end of each of n periods at periodic discount rate r. Show every algebraic step.

B4. **(6 marks)** You wish to accumulate Rs. 50,00,000 in 25 years by making equal end-of-year deposits into an account earning 7.5% p.a. compounded annually. What must each annual deposit be? *(Hint: this is a future-value-of-annuity problem – derive the FV-of-annuity formula from the PV-of-annuity formula you already know, by compounding forward.)*

B5. **(6 marks)** A bond has face value Rs. 1000, an annual coupon rate of 6.5%, and 5 years to maturity. If the market's required yield is 7.2%, compute the bond's fair price. Is the bond trading at a premium or a discount to face value? Explain why, referencing the relationship between the coupon rate and the yield.

B6. **(5 marks)** **[Python]** Write a short Python function `bond_price(face, coupon_rate, n, y)` that returns the price of an annual-coupon bond, and use it to reproduce your answer to B5. Then plot the bond's price as a function of yield y ranging from 2% to 12%, and comment on the shape of the curve you obtain (is it a straight line? why or why not?).

Part C – Yield Curves (30 marks)

C1. **(4 marks)** Explain, in 3–4 sentences and without formulas, why a single interest rate is insufficient to describe a government bond market, and what object we use instead.

C2. **(4 marks)** Define, in one sentence each, a *spot rate*, a *par yield*, and a *forward rate*. State which of the three is *directly observed* in the market and which two must be derived.

C3. **(6 marks)** Given the following annual par yields for a hypothetical government bond market (all bonds priced at par, annual coupons):

Maturity (yrs)	1	2	3	4
Par yield	5.20%	5.45%	5.60%	5.70%

Hand-bootstrap the zero rates $z(1), z(2), z(3), z(4)$, showing every step of your working (algebraic, not numerical/iterative methods – this should be solvable in closed form at each step, exactly as done in Lecture 3).

C4. **(4 marks)** Using your answer to C3, compute the 1-year forward rate starting in 3 years, $f(3,4)$. Is it larger or smaller than $z(4)$? Is this consistent with the shape of the input par curve? Explain.

C5. (4 marks) A friend argues: "Bootstrapping is unnecessary – I'll just use each bond's YTM directly as if it were the zero rate for that maturity, it's much simpler." Explain, using a specific consequence, why this shortcut introduces a systematic error, and under what special condition (on the shape of the curve) the error would vanish.

C6. (8 marks) [Python] Extend the bootstrapper you built in this week's Lab (or write a fresh, self-contained script) to bootstrap the zero curve for the C3 data set above, and confirm your hand calculations match your code's output to at least 4 decimal places. Then use your bootstrapped curve to price a new, off-benchmark bond: face value Rs. 100, coupon 6%, maturity 3.5 years *(treat this as paying its final coupon and face value together at $t = 3.5$, with ordinary annual coupons at $t = 1, 2, 3$ before that – you will need to interpolate your zero curve to get $z(3.5)$)*. Report the price and briefly comment on whether it looks reasonable given the shape of the curve.

Part D – Integrative Question (10 marks)

D1. (10 marks) An insurance company sells a policy today that promises the policyholder a single lump-sum payout of Rs. 20,00,000 exactly 10 years from now. The company wants to fund this liability today by purchasing a portfolio of government bonds from the C3 market (maturities 1–4 years only are available for this question – assume any maturities beyond 4 years are unavailable). Explain, using ideas from *all three* lectures this week (fixed vs. residual claims, discounting, and the term structure), why the insurer cannot fully "match" this liability using only the instruments available in this market, and describe qualitatively (no calculation required) what risk the insurer is left exposed to as a result.

End of Assignment. Solutions will be released after the submission deadline.